Gravity

THE FORCE OF GRAVITY AFFECTS EVERY OBJECT IN THE UNIVERSE.

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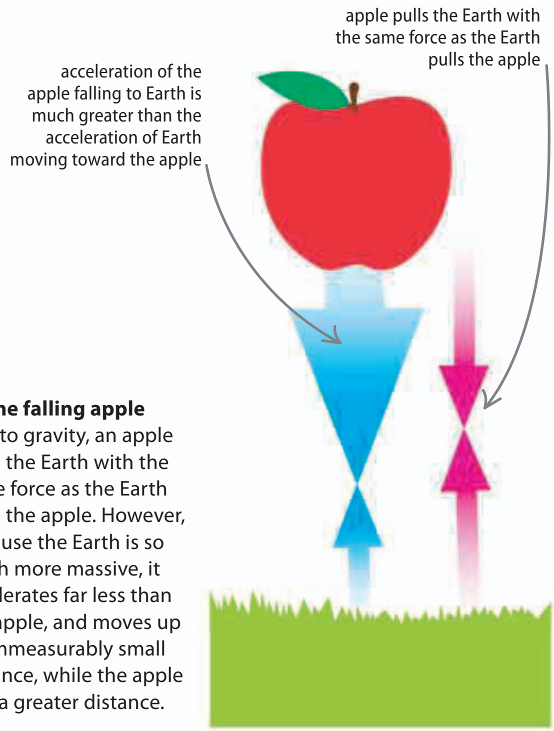
Gravity is the force that holds planets together and keeps them orbiting stars, as well as holding us to the Earth.

Attraction

Gravity is a force of attraction. Although all objects attract all others, gravity between objects on Earth is usually too small to notice. This is why it took a genius like Isaac Newton to understand that gravity does affect all objects, whatever their size.

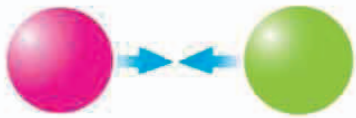


▷ **The falling apple**
Due to gravity, an apple pulls the Earth with the same force as the Earth pulls the apple. However, because the Earth is so much more massive, it accelerates far less than the apple, and moves up an immeasurably small distance, while the apple falls a greater distance.

Physicists think that the force of gravity is carried by **tiny particles called gravitons**—but they have yet to find any.

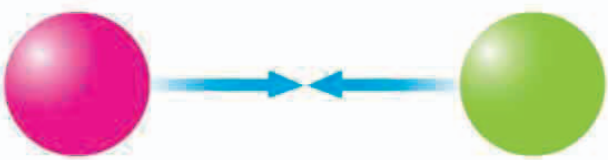
Universal law

Isaac Newton discovered that gravity affects everything in the Universe. He explained that the gravitational force between two objects, such as planets, depends on their masses and the distance between them. Newton also showed that spherical objects, such as the Earth, act as if all their mass is concentrated at their centers.



△ **Attraction**
All objects are attracted to each other by the force of gravity. Above, two objects of the same mass are attracted by the pull of this gravitational force.

△ **Double the mass**
Changing the masses of the two objects so that they are twice as heavy will make the gravitational force between them four times as strong.



△ **Increase the distance**
Changing the distance between the two objects so that they are twice as far away from each other will make the gravitational force four times less.